

Fixed Points VI: Formal Synthesis and Physical Interpretations

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Part VI of VI in the Fixed Point series. As both the cornerstone of the Modal Triplet Theory (MTT) collection and a stand-alone development, the series is intended to function simultaneously as a basis and as a self-contained study. Each paper in the series builds upon its predecessors, extending the fixed-point framework step by step.

Abstract

We synthesize the corrected FP–I–V results without promoting their control parameter, projections, or diagnostics into an unproved fundamental field theory. The rigorous spine consists of conditional projected fixed-point existence, strict-Lyapunov promotion to equilibrium, joint-mode damping and disturbance floors, perturbative persistence of a curved spectral cluster, curvature leakage, intrinsic first-order centroid modulation, frozen linear Ornstein–Uhlenbeck covariance, canonical-correlation bounds, and declared admissibility exits. We distinguish three logical levels throughout: inherited theorem, conditional model completion, and physical interpretation. A Lorentzian gauge/gravity action, quantum covariance, particle identities, merger, measurement, and cosmology belong to the latter two levels unless additional source and equivalence theorems are supplied. We also correct the Gaussian Lyapunov sign and nonnormal resolvent bounds, and show why an instantaneous equal-time bilocal kernel is not microcausal merely because it leaves the principal symbol unchanged. A local mediator is the consistent route to causal overlap dynamics. A separate, explicitly non-FP ledger records later scoped q79 geometry, quantum-operation, and QFT advances without retroactively promoting them to conclusions of the fixed-point series.

Revision note for this edition

Supersedes. *Fixed Points VI: Formal Synthesis and Physical Interpretations*, version 4.

Reason. Version 4 correctly narrowed the FP series, but it predates the exact q79 lane-factor, projective-module, connection-compiler and nonlinear strain-quotient results, as well as later scoped QM and QFT closures. Without a separate status layer, “not derived by FP” could be misread as “not achieved anywhere downstream.”

Resolution. Version 5 preserves the FP theorem boundary, corrects the q79 preprojection types, and adds an independently sourced downstream ledger with exact closure counts and nonpromotion clauses.

Retained result. The complete FP–I–V fixed-point, damping, curvature, covariance, and admissibility spine survives at its declared theorem tier.

Remaining boundary. Selected HYM endpoints and action, the physical q79 Hessian and reduction, universal apparatus and actualization, a fixed-coupling continuum QFT, selected global state, RG and matching, and cosmology remain downstream programs.

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1 Logical levels and common geometry

Every statement in this synthesis has one of the following statuses.

Inherited theorem. A consequence of FP I–V under hypotheses restated or cited here.

Conditional completion. A mathematically specified extra model whose conclusions hold if that model and its hypotheses are adopted.

Physical interpretation. A proposed reading that is not established by the fixed-point theorems alone.

Let $\mathcal{H}$ be the declared control Hilbert space. The joint internal operator is constructed from three declared vertical operators using the strong-commutation and domain hypotheses of FP II. Circle–lens–nil labels refer here to obstruction, rank, or operator data; they do not assert a literal product or nesting of three manifolds. In the current q79 specialization the ranks 1, 2, and 3 act on an external lane tensor factor, not inside an irreducible stable HYM gauge bundle. The shared differential line is a separate flat scalar tensor factor. The accepted 27-state algebra is post-projection source data, not a rank-27 Galerkin subspace of the rank-102 physical deformation carrier. The six-coordinate strain carrier is a nonlinear orientation-forgetting quotient shadow, not a linear rank-six subspace of $\text{Herm}(3)$ [8]. The auxiliary $L(3, 1) \times \text{Nil}_3$ model is not identified with the q79/Fu–Yau compactification. The shared circle is common $U(1)$ phase/holonomy data and is counted once.

The coherent projector P is the joint spectral projector. A product of three bundle projectors is permitted only when those projectors strongly commute; otherwise the product need not be an orthogonal projector. Write $Q = I - P$.

On a curved bundle, the normalized Laplace-type operator is

$$L_x = \nabla_x^* \nabla_x + \mathcal{R}_x.$$

When the selected low cluster remains separated, FP IV defines the curved Riesz projector $P_{\mathcal{R}}(x)$. Retaining the old projector P instead requires the off-diagonal leakage block $Q\mathcal{R}P$.

The parameter of the FP gradient flow is a stabilization or control parameter. Identifying it with Lorentzian proper time, coordinate time, or a quantum evolution parameter is a conditional completion, not an inherited theorem.

2 Fixed-point and equilibrium synthesis

Let Φ_τ be a well-defined time- τ map for the declared control flow and set $T = P\Phi_\tau$ on a closed bounded convex set $K \subset \mathcal{H}$.

Theorem 2.1 (Corrected fixed-point chain). *Assume one of the following FP I alternatives:*

1. $T(K) \subset K$ and $T : K \rightarrow K$ is continuous and compact; or
2. $T(K) \subset K$ and T is continuous and condensing for a declared measure of noncompactness.

Then T has a fixed point $u_ \in K$. This proves $P\Phi_\tau(u_*) = u_*$. If, in addition, the trajectory through u_* remains in the projected invariant set and obeys a strict Lyapunov identity*

$$\mathcal{C}(u(\tau)) - \mathcal{C}(u(0)) = - \int_0^\tau \mathcal{D}(u(s)) ds, \quad \mathcal{D} \geq 0,$$

with $\mathcal{D}(u) = 0$ exactly at equilibria, then u_ is an equilibrium of the declared flow.*

Proof. The first conclusion is Schauder's or Darbo–Sadovskii's theorem. At a projected recurrent point, the Lyapunov values at the two endpoints agree. The displayed identity forces $\mathcal{D} = 0$ along the intervening orbit, and strictness gives equilibrium. Without the second step, a projected time-step fixed point is not automatically a steady solution. $\square$

Proposition 2.2 (Conditional contraction uniqueness). *Suppose a steady equation can be written on a specified Banach space as $u = -A^{-1}N(u)$, where A^{-1} exists on that space and*

$$\|A^{-1}\| \text{Lip}(N) < 1$$

on an invariant complete subset. Then the steady solution there is unique. No uniqueness conclusion follows from a vertical spectral gap if A has an unremoved kernel or if the displayed inverse estimate is not proved.

3 Joint-mode stability and disturbance floors

Let α be one joint modal label, including multiplicity. After the one-sided nonlinear estimate, suppose a noncoherent amplitude satisfies

$$\frac{d}{dt}|a_\alpha| \leq -\gamma_\alpha|a_\alpha| + |f_\alpha(t)|, \quad \gamma_\alpha > 0.$$

Then the deterministic input-to-state floor is $\|f_\alpha\|_\infty/\gamma_\alpha$. For the exact scalar Itô equation

$$da_\alpha = -\gamma_\alpha a_\alpha dt + \sqrt{q_\alpha} dW_\alpha,$$

the stationary variance is $q_\alpha/(2\gamma_\alpha)$. Deterministic force amplitude and stochastic power have different units and cannot be represented by one shared disturbance threshold.

For infinitely many modes, bundlewise stochastic control requires the appropriate weighted covariance trace; deterministic control requires its own weighted series. These are sufficient nonlinear stability criteria. Gaussian invariant laws are asserted only for exact linear OU systems.

4 Curvature, centroid motion, and interaction

Let the uncurved noncoherent spectrum start at $\lambda_* > 0$. If $\|\mathcal{R}\| < \lambda_*/2$, the selected low spectral cluster remains separated and has a curved Riesz projector of the same rank. If the uncurved projector is retained and the one-sided Q -sector damping margin is $\gamma_Q > 0$, FP IV gives the schematic sharp form

$$\frac{d}{dt}\|q\| \leq -\gamma_Q\|q\| + \|Q\mathcal{R}P\|\|p\| + \|QF\|,$$

and therefore a curvature-induced leakage floor.

For a localized profile inside a strongly convex normal ball, the centroid is the Karcher mean. A first-order gradient parent flow yields a first-order modulation equation

$$G_{ij}(X)\dot{X}^j = -\partial_i V_{\text{eff}}(X) + \varepsilon_i.$$

A Newton equation requires a separately declared inertial parent theory.

For two profiles, absolute cross-term control proves only

$$|E_{\text{int}}| \leq C\mathcal{O}.$$

Attraction requires $E_{\text{int}} \leq -c\mathcal{O}$; repulsion requires the opposite sign. Even a proved attractive sign does not establish merger, collapse, or a transition rate without a dynamical connection theorem.

5 Frozen Gaussian multi-structure theory

Consider the real linear SDE

$$d\zeta = A\zeta dt + B dW_t, \quad Q := BB^\top. \quad (5.1)$$

Theorem 5.1 (Semigroup covariance and resolvent bounds). *Assume*

$$\|e^{tA}\| \leq Me^{-\omega t}, \quad t \geq 0, \text{quad}M \geq 1, \text{quad}\omega > 0.$$

Then (5.1) has stationary covariance

$$\Sigma = \int_0^\infty e^{tA} Q e^{tA^\top} dt,$$

which solves

$$A\Sigma + \Sigma A^\top + Q = 0.$$

Moreover,

$$\|\Sigma\| \leq \frac{M^2\|Q\|}{2\omega}, \quad \|A^{-1}\| \leq \frac{M}{\omega}.$$

Proof. The semigroup estimate makes both integrals converge. Differentiation of the covariance integrand gives the Lyapunov equation, while $A^{-1} = -\int_0^\infty e^{tA} dt$. Taking norms proves the bounds. $\square$

For a normal A , one may often take $M = 1$ with ω given by the spectral abscissa. For a nonnormal matrix, the spectral abscissa alone does not imply $\|A^{-1}\| \leq 1/\omega$ or $\|\Sigma\| \leq \|Q\|/(2\omega)$; transient amplification must be controlled.

Under the block-diagonal damping assumptions of FP V, the cross covariance satisfies a Sylvester equation. Canonical correlation is normalized by the smallest positive eigenvalues of the marginal covariance blocks, not their largest operator norms.

6 Classical covariance versus quantum covariance

The covariance produced by (5.1) is classical. To interpret it as a bosonic quantum covariance one must separately specify canonical commutation relations, a value of $\hbar$, and quantum dynamics whose noise and damping satisfy the complete-positivity constraints. Only then is

$$\Sigma + \frac{i\hbar}{2}\Omega \succeq 0$$

the physical uncertainty condition. Classical positivity of Σ does not imply this inequality.

For a declared bipartite Gaussian quantum state, nonpositivity after partial transpose certifies entanglement. PPT is exact for the two-mode 1×1 case; in general multimode bipartitions PPT need not be sufficient for separability. These facts classify an already constructed quantum Gaussian state. They do not quantize the FP control flow or derive particle statistics.

7 Admissibility and exit

Let $m_j(x)$, $j \in \mathcal{J}$, be a finite family of continuous declared margins: curved cluster separation, one-sided damping, or explicit deterministic/stochastic performance tolerances. Define

$$\mathcal{D}_\varepsilon = \{x : m_j(x) \geq \varepsilon \ \forall j\}, \quad \mathfrak{B}(x) = \max_j [-m_j(x)].$$

Then $x \in \mathcal{D}_0$ exactly when $\mathfrak{B}(x) \leq 0$. Along a continuous path, a change from $\mathfrak{B} \leq 0$ to $\mathfrak{B} > 0$ has a first boundary time at which $\mathfrak{B} = 0$.

This is an exact exit diagnostic for the declared constraints. It does not prove that $\mathfrak{B}$ is a physical energy or force. It does not establish which state follows exit. A mountain-pass theorem needs an actual energy functional, and basin selection needs a post-exit evolution or transition kernel.

For affine observables of a frozen Gaussian system, FP V supplies finite-grid and Borell–TIS continuous-time exit estimates. A Lipschitz nonlinear function of a Gaussian process is not generally Gaussian. Equal-time cross-correlation can bound simultaneous observed exits, but cannot prove causal propagation or non-propagation.

8 Status of a Lorentzian master action

A local Lorentzian gauge/gravity/matter action may be appended as a *conditional completion*, for example schematically

$$S_{\text{cand}} = \int \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \Lambda - \frac{1}{4} \sum_r \text{tr } F_r^2 - \sum_a |D\varphi_a|^2 - V(\varphi) + \sum_b \bar{\psi}_b (i\gamma^\mu D_\mu - M_b) \psi_b \right] d^4x.$$

The FP I–V theorems do not select the gauge group, representations, number of fields, couplings, Yukawa matrices, or potential. Nor do they prove that the gradient control flow is the Euler–Lagrange evolution of S_{cand} .

If a mass term depends on curvature, such as $M_a^2 = m_a^2 + \xi_a R$, its metric variation contributes nonminimal terms to the gravitational field equation; it cannot be inserted while retaining the minimally coupled Einstein equation unchanged. Likewise, a point-particle Lorentz-force law requires a controlled localized-solution limit and is not a consequence of gauge covariance alone.

Three internal structures do not by themselves prove three gauge factors, bosonic or fermionic statistics, bifundamental matter, the Standard Model particle assignment, anomaly cancellation, or equivalence with a quantum field theory. Those require independent representation, source, and matching theorems.

9 Independent downstream q79 status

This section is a status reconciliation against the canonically verified q79 source repository at commit 1615da7 [8]. None of the items below is presented as a consequence of the FP I–VI theorems.

Geometry-to-operator bridge. For a supplied finite-rank Hermitian bundle and unitary connection, exact projective-module connection, curvature, coupled-differential and functorial Hessian transport is closed at 10/10. The explicit Čech projector plus connection-correction compiler is closed at 12/12, and the intrinsic relative-phase quotient with reduced-Green shorted Hessian is closed at 14/14. These are universal or conditional constructions. The actual same-source q79 instantiation remains 0/3: selected visible/hidden HYM endpoints, their physical action-derived Hessian, and an intrinsic finite subspace with either vanishing compression residual or an executed Feshbach identity are not yet supplied.

Canonical quantum operation. The selected q79 binary one-anchor recorder has an exact commuting output algebra, stopped-output measure, and quadratic P/Q probabilities for every declared preparation in that apparatus context. This closes the operational Born-output tier without a new stochastic primitive, fit, or observed probability. It does not prove a source rule for every apparatus, derive probability semantics before the normal-state formalism, or select one ontically actual history.

Quantum field theory. The framed free massless Dirac even-observable net is closed at its selected q79 tier. Independently, the selected finite SM carrier has been composed through classical BV, gauge-fixed Green-hyperbolic/equicausal structure, a formal anomaly-free all-orders QME, formal positive physical state spaces, and presentation transport. A

literal physical C-star reference net is closed at $\lambda = 0$. At nonzero coupling, an actual finite-regulator C-star landing is closed at 5/5, while only the normalized same-source first tangent is accepted in the physical continuum-promotion table (1/9). The finite spectral chiral measure is closed on each certified connected gapped presentation orbit (7/7), but its selected full-domain extension remains 0/4. Fixed-coupling continuum convergence and uniqueness, a selected global interacting state, numerical RG/matching, and nonperturbative completion remain open.

10 Bilocal overlap and causality

Consider an equal-parameter interaction of the form

$$\mathcal{N}[\phi](t, x) = \int_{\Sigma_t} K(x, y) |\phi(t, y)|^2 d\mu_t(y) \phi(t, x).$$

Proposition 10.1 (Instantaneous bilocal obstruction). *If $K(x, y)$ is nonzero for spatially separated x and y , the equation at (t, x) depends instantaneously on data at y . Leaving the differential principal symbol unchanged is therefore insufficient to prove the usual local domain-of-dependence property. An L^1 bound on K may help local well-posedness, but it does not establish microcausality.*

Proof. Choose two initial data sets that agree near x but differ near a separated point y where $K(x, y) \neq 0$. Their bilocal source values at (t, x) differ at the same parameter time. Thus the source is not determined by data in an arbitrarily small causal neighborhood of x . $\square$

Proposition 10.2 (Local-mediator completion). *Introduce a local mediator χ with a hyperbolic equation, for example*

$$(\square_g + m_\chi^2)\chi = g_\chi |\phi|^2,$$

and couple χ locally back to ϕ . Under the standard regularity, gauge, and constraint hypotheses for the resulting local hyperbolic system, its domain of dependence is governed by the local characteristics. Eliminating χ with a retarded Green operator produces a causal retarded memory kernel, not in general a symmetric equal-time bilocal action.

This local parent theory is the preferred route if overlap dynamics is meant to represent physical causal interaction.

11 Physical interpretation ledger

Supported within the declared control model. The declared model supports projection into coherent and noncoherent sectors under operator-domain hypotheses. It also supports conditional fixed points, strict-Lyapunov equilibrium promotion, damping estimates and disturbance floors, perturbative curved-cluster persistence, curvature leakage, intrinsic first-order centroid modulation, frozen linear covariance, and admissibility exit diagnostics.

Conditional after adding a model. Gauge forces, Lorentzian motion, quantum uncertainty, Gaussian entanglement, decoherence, thermodynamic entropy production, and causal overlap can be studied after a compatible local Lorentzian or open-quantum model is specified. Their parameters and consistency conditions are additional data.

Not derived by this series. The present theorems do not derive boson/fermion identity, particle masses, Standard Model representations, merger or collapse rules, the Born rule, measurement outcomes, emergent time, inflation, horizons, or a cosmological arrow of time. Section 9 records independent scoped results on some of these bridges; their existence does not make them conclusions of FP I–VI.

12 Synthesis theorem and remaining obligations

Theorem 12.1 (Scoped FP synthesis). *Assume the operator, compactness or condensing, strict-Lyapunov, one-sided damping, curvature-smallness, covariance, and margin hypotheses stated in FP I–V and summarized above. Then the declared control model has:*

1. *a projected time-step fixed point, promoted to an equilibrium only by the strict Lyapunov condition;*
2. *quantified noncoherent damping and deterministic/stochastic floors;*
3. *a persistent curved low cluster and a quantified old-projector leakage term;*
4. *intrinsic first-order centroid modulation under the modulation hypotheses;*
5. *exact stationary covariance for the frozen linear OU sector and correctly normalized correlation bounds; and*
6. *exact detection, plus conditional Gaussian probability bounds, for exit from a declared admissible domain.*

No Lorentzian, quantum, particle-physics, merger, measurement, or cosmological claim follows without its additional completion theorem.

The principal remaining obligations for a physical theory are therefore constructive rather than interpretive: emit the selected visible/hidden HYM endpoints; derive the upper action and physical rank-102 Hessian; execute its invariant finite restriction or Feshbach/shorted reduction; extend the canonical quantum operation beyond one binary apparatus if that stronger claim is intended; construct the fixed-nonzero-coupling continuum QFT and a selected global state; prove the relation between physical evolution and the FP control flow; and complete RG, uncertainty, and empirical matching without importing target observables as source data.

13 Conclusion

The corrected Fixed Points series is a coherent conditional control and spectral framework. Its strongest results concern existence under explicit compactness hypotheses, stability margins, curved spectral persistence, leakage, modulation, Gaussian covariance, and admissibility diagnostics. FP VI does not weaken those results by surrounding them with broader interpretations; it identifies exactly what they prove and exactly which bridges remain before they can support a fundamental spacetime, quantum, or particle theory.

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